# Solving NP-Hard Problems Using Simulated Annealing

## ADDIS ABABA UNIVERSITY | Assignment 1 Submission

Registered algorithm: Simulated Annealing (SA)

Problem: Traveling Salesperson Problem (TSP) — att48, ch130, a280

Group: 5 | Repository URL: [INSERT PRIVATE GITHUB URL]

Member roster: smn | [INSERT STUDENT ID] | [INSERT TASK] | [INSERT %]

Add remaining members and ensure contributions sum to 100%.

## Executive summary

This study implements simulated annealing from scratch for permutation-encoded TSP tours. It compares SA with an equal-budget random-restart 2-opt baseline over 30 independent runs per instance. The objective is tour length minimization.

## Submission checklist

Code, data scripts, experiments, raw CSV traces, plots, and reproducibility commands are in the repository. Complete the bracketed cover-page fields before submission.

# Natural Inspiration and Mathematical Formulation

## Annealing analogy

Metallurgical annealing heats a material so particles move freely, then cools it slowly so the material settles into a low-energy structure. In SA, high temperature enables exploration by accepting some worse tours; low temperature emphasizes exploitation and improvement.

## Metropolis rule and cooling

For cost change Δ=f(candidate)-f(current), accept every improving move. For a worsening move, accept with P(accept)=exp(-Δ/T). Temperature is updated once per block: T_new=T_old*r, where r=(T_final/T_initial)^(1/number_of_blocks). Initial temperature is estimated from the median worsening 2-opt change d using T0=-d/log(target_acceptance), then multiplied by the tuned temperature scale.

## Encoding and 2-opt

A solution is a permutation of city indices, so every city appears exactly once. A 2-opt move reverses a contiguous segment and therefore preserves feasibility. With edges (a,b) and (c,d) replaced by (a,c) and (b,d), Δ = dist(a,c)+dist(b,d)-dist(a,b)-dist(c,d).

## Numerical example

Suppose current cost is 120, candidate cost is 130, so Δ=10. At T=20, P=exp(-10/20)=0.6065. If the random draw is 0.40, accept the worse candidate because 0.40<0.6065. If the draw is 0.80, reject it. This temporary worsening can escape a local optimum.

# Problem and Experimental Setup

## TSP formulation and benchmarks

Given cities and distances d(i,j), minimize the closed-tour length Σ d(t_i,t_(i+1) mod n), subject to t being a permutation. att48 has 48 cities and ATT distances with BKS 10628; ch130 has 130 cities and EUC_2D distances with BKS 6110; a280 has 280 cities and EUC_2D distances with BKS 2579. BKS values are from the TSPLIB solutions file and Reinelt (1991).

## Baseline and evaluation budget

The baseline is random-restart improving 2-opt. It accepts only improving moves and restarts after a fixed number of non-improving attempts. Both methods use 50,000 objective evaluations per run; the initial tour cost counts as evaluation 1. Each method uses 30 runs with seeds 2026–2055.

## Parameter tuning

The tuning grid uses five runs per configuration and seeds beginning at 9000. It tests target acceptance, initial-temperature scale, final temperature, and moves per temperature block. The selected settings are att48 (0.7, 0.75, 0.1, 200), ch130 (0.3, 0.25, 0.5, 50), and a280 (0.1, 0.25, 0.5, 200).

## Environment

Python 3.12.3, matplotlib 3.6.3, scipy 1.11.4, Linux x86_64. Record the exact CPU and RAM before submission.

# Results and Convergence Analysis

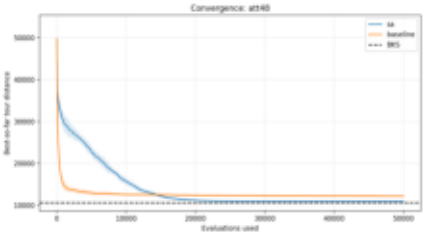
## 30-run summary (mean gap is relative to published BKS)

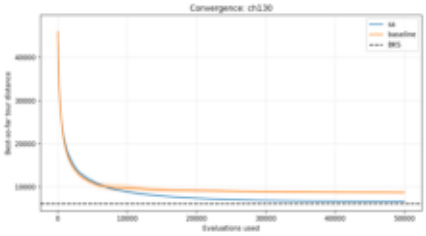
```
SA       att48  best=10638.0 mean=10902.87 median=10864.0 std=158.61 gap=2.59%
SA       ch130  best=6419.0 mean=6662.27 median=6657.0 std=152.51 gap=9.04%
SA       a280   best=3587.0 mean=3782.00 median=3788.0 std=100.50 gap=46.65%
Baseline att48  best=11514.0 mean=12144.07 median=12135.0 std=275.31 gap=14.26%
Baseline ch130  best=7945.0 mean=8708.00 median=8731.5 std=322.15 gap=42.52%
Baseline a280   best=4227.0 mean=4539.00 median=4498.5 std=193.22 gap=76.00%
```

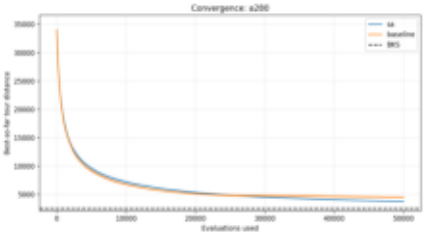
## Statistical comparison

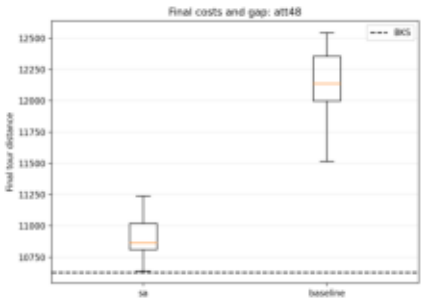
Two-sided Mann–Whitney U at α=0.05: a280 p=3.01e-11, att48 p=3.02e-11, ch130 p=3.01e-11. SA medians are lower for a

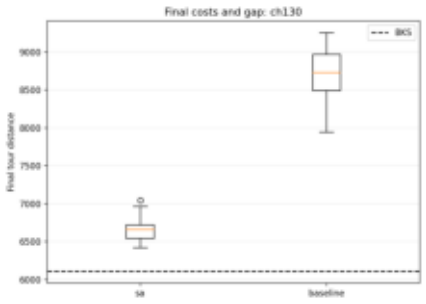
## Convergence and final-cost plots

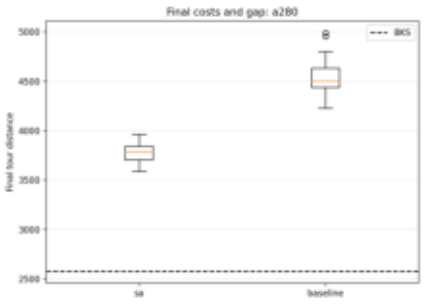


# Discussion, Limitations and References

## Discussion

SA balances exploration and exploitation through the temperature schedule. The heat maps document sensitivity to acceptance targets, initial temperature, final temperature, and block length. The a280 mean gap of 46.65% shows a scalability limitation: 280-city tours have a larger and more difficult search space, while the fixed 50,000-evaluation budget and simple 2-opt neighborhood limit refinement.

## Limitations

The tuning study uses five runs per configuration, so its estimates are noisy. Only one neighborhood and one baseline are used. Hardware details must be completed from the actual machine, and the cover-page identities must be completed by the group.

## References

[1] S. Kirkpatrick, C. D. Gelatt Jr., and M. P. Vecchi, “Optimization by Simulated Annealing,” Science, vol. 220, no. 4598, pp. 671–680, 1983.

[2] N. Metropolis et al., “Equation of State Calculations by Fast Computing Machines,” Journal of Chemical Physics, vol. 21, no. 6, pp. 1087–1092, 1953.

[3] G. Reinelt, “TSPLIB—A Traveling Salesman Problem Library,” ORSA Journal on Computing, vol. 3, no. 4, pp. 376–384, 1991.

[4] G. A. Croes, “A Method for Solving Traveling-Salesman Problems,” Operations Research, vol. 6, no. 6, pp. 791–812, 1958.